

Characterizing Query Planners in Heterogeneous Clusters

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Abstract

Work in computing research has long relied on assumptions we test directly here. Neither of the alternative formulations improved on the simple additive model in our setting. We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses.

1. Introduction

Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses. The measurements reported here were collected over twelve independent trials under controlled conditions. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Neither of the alternative formulations improved on the simple additive model in our setting. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

Replicate agreement was high, suggesting the protocol is robust to small handling differences. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Taken together, these findings argue for treating the two regimes separately in future analyses. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. We release the processing scripts and intermediate artefacts to support independent verification. Longer observation windows narrow the confidence interval but do not shift the central estimate. Sensitivity to the smoothing window proved modest across the range we examined.

Calibration drift over the trial period remained below the threshold set out in the protocol. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost. Residual error grows approximately linearly with the number of coupled parameters.

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2. Related Work

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3. Materials and Methods

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4. Experimental Setup

Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. Longer observation windows narrow the confidence interval but do not shift the central estimate. Calibration drift over the trial period remained below the threshold set out in the protocol. Sensitivity to the smoothing window proved modest across the range we examined. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. We release the processing scripts and intermediate artefacts to support independent verification.

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5. Results

The measurements reported here were collected over twelve independent trials under controlled conditions. The reduced model retains most of the explanatory power at a fraction of the computational cost. We release the processing scripts and intermediate artefacts to support independent verification. Neither of the alternative formulations improved on the simple additive model in our setting. Sensitivity to the smoothing window proved modest across the range we examined.

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6. Discussion

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Calibration drift over the trial period remained below the threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification. Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The measurements reported here were collected over twelve independent trials under controlled conditions. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting.

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7. Limitations

Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Taken together, these findings argue for treating the two regimes separately in future analyses. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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8. Conclusion

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